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*Project title:* ***Priority Locations for Green Space Interventions in Greater Manchester***

# 1. Introduction

Urban green space is essential for supporting public health, environmental quality, and social well-being, particularly in densely populated cities. A wide range of studies shows that having convenient access to good quality green infrastructure is linked to better physical and mental health, reduced stress, and lower health inequalities with especially strong benefits for vulnerable groups such as children and people living in deprived neighbourhoods (Dennis et al., 2016; Dennis et al., 2020). However in many UK cities green space is still distributed unevenly and often lines up with existing patterns of socio-economic disadvantage and environmental inequality. This gap means that national goals for fairer green space access are tough to turn into practical, location-specific actions backed by solid evidence and pose a major challenge for urban planners.

Greater Manchester offers a particularly relevant case study for exploring these challenges. Greater Manchester, being a big post-industrial city-region with huge contrasts between its inner city spots, old factory areas, and suburbs, shows really clear spatial differences in green space access, how healthy the vegetation looks and socio-demographic vulnerability levels. But putting these standards to effective use at the neighbourhood level means looking at several spatial factors together like physical access to green spaces, the condition of the vegetation, levels of socio-economic deprivation, and characteristics of the local population (Natural England, 2010). GIS tools, especially GIS based multi-criteria decision analysis (MCDA), provide a solid way to handle this complexity by letting you pull together different datasets into one clear analytical model (Malczewski, 2006;). This study therefore aims to identify priority spots for green space improvements across Greater Manchester, tackling this research question:

**Where within Greater Manchester should green space interventions be most effectively targeted to maximise environmental and social benefits, particularly for vulnerable populations, based on integrated spatial evidence?**

# 2. Methodology

This study uses GIS-based multi-criteria suitability analysis to pinpoint Greater Manchester neighbourhoods where green space improvements are most urgently needed, based on overlapping environmental and social factors. I opted this approach because green infrastructure planning is so multi-faceted that you have to consider access, vegetation quality, socio-economic conditions, and population vulnerability all at the same time, rather than just looking at single indicators (Malczewski, 2006, Jankowski, 1995). Combining those criteria into one analytical framework makes the whole process transparent and gives a solid, defensible way to prioritise intervention areas.

This analysis focuses the Greater Manchester city region at the Lower Layer Super Output Area (LSOA) scale, which is a standard neighbourhood-level unit commonly used in UK planning, deprivation studies, and public health research. This spatial scale lets me zoom in on local conditions in enough detail while still being spot-on for policy and decision-making needs. Four key datasets were used to get at the main aspects of green space need: public

greenspace data (OS Open Greenspace), vegetation condition from Sentinel-2 images via NDVI, socio-economic deprivation (IMD 2019) and child population at LSOA level. All datasets were aligned to the same spatial setup so they could be compared fairly and kept reliable when overlaid together. Each dataset was turned into a criterion showing relative shortage or priority, with everything set up so higher values mean greater need for green space fixes. Greenspace provision was measured as an accessibility gap basically, the further people are from public green areas, the bigger the need which lines up with UK planning guidance on how close access should be (Natural England, 2010). Vegetation condition was represented using an NDVI-based measure, flipped around to highlight areas with poorer vegetation rather than just greenness. Socio-economic vulnerability came from normalised IMD scores and child population was added as a key priority factor since evidence shows kids get especially strong long-term gains from good green spaces (Dennis et al., 2016; Dennis et al., 2020).

All criteria were normalised to a 0–1 scale to allow smooth integration from different data sources, ensuring no variable dominated due to its range or units. Using a weighted linear method, the criteria were combined into a continuous suitability map, with higher weights assigned to socio-economic deprivation and child population to emphasise equity, while greenspace access and vegetation condition served as supporting factors.

The final map highlights neighbourhoods where several environmental and social challenges overlap, indicating where green space improvements could have the biggest impact. For easier interpretation, the suitability values were grouped into five categories, ranging from very low to very high, with the highest class identifying key priority areas. Scores were aggregated at the LSOA level and compared with child population patterns to identify neighbourhoods where environmental needs coincide most strongly with potential social benefits, forming a robust evidence base for green space planning across Greater Manchester.

# 3. Results & Discussion

## 3.1 Environmental Deficits Across Greater Manchester

Baseline maps of greenspace access and vegetation cover provide the environmental context for the suitability model; they describe inequalities in provision rather than priority intervention sites. The greenspace access deficit (Figure 3.1) is highest in dense inner-urban areas especially central Manchester and parts of Salford, Oldham, and Rochdale and generally lower in suburban and surrounding neighbourhoods. The NDVI-derived vegetation deficit (Figure 3.2) shows a related but different pattern, with the strongest vegetation shortfalls in the urban core and along major transport corridors, and greener conditions towards suburban and peri-urban areas.

Importantly, these gaps don’t always align perfectly. Some areas lack nearby formal greenspace even with decent vegetation around them, while others face both poor access and low greenness. This shows that "need" is truly multi-dimensional, which backs up the value of combining these indicators into the overall suitability analysis that comes next.

*Figure 3.1: Distance-based greenspace access deficit across Greater Manchester.*

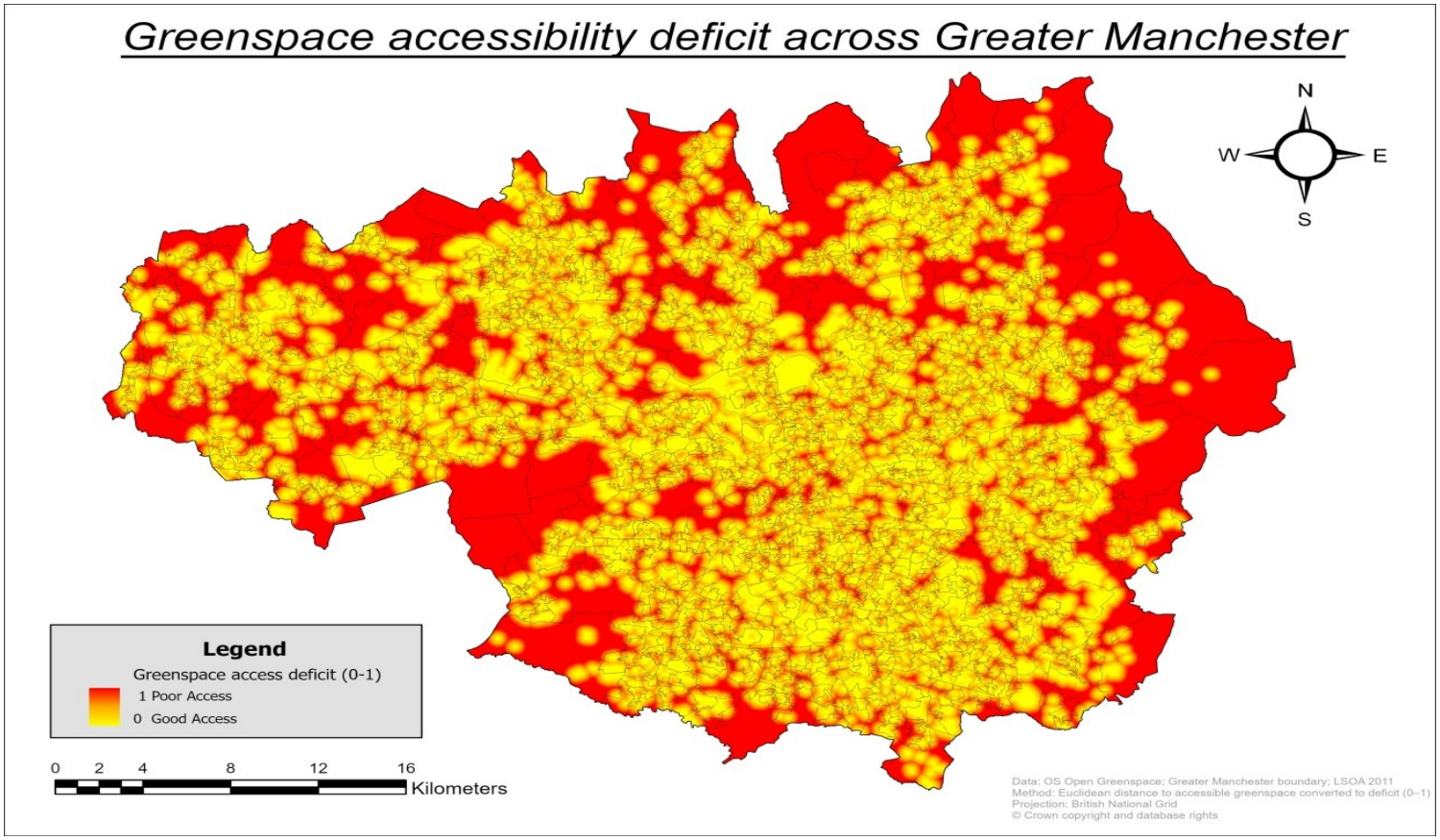
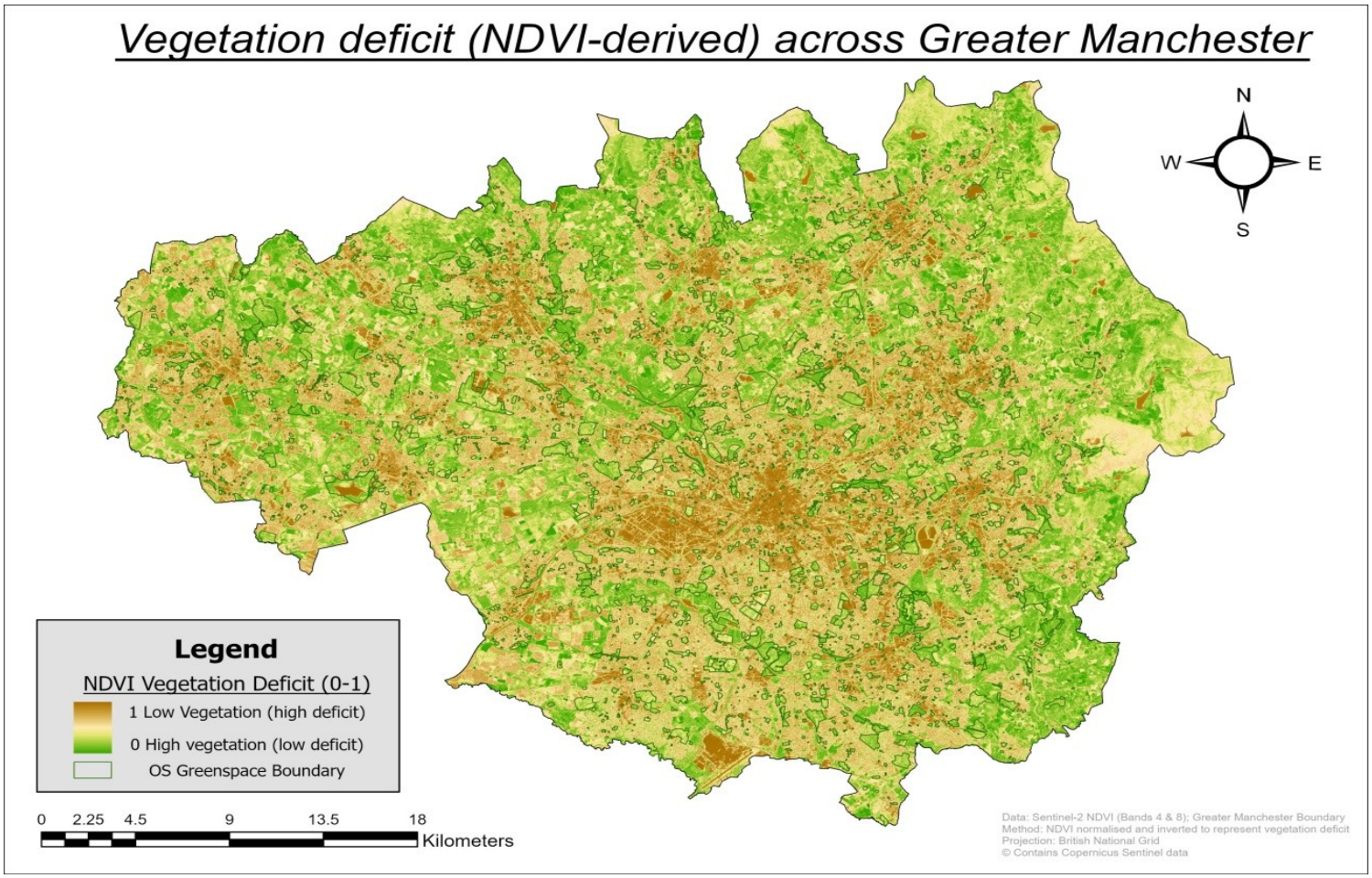



***Figure 3.2 (NDVI-derived vegetation deficit) presents a related but distinct pattern***



## 3.2 Composite Suitability Modelling Results

The composite suitability model integrates environmental and socio-demographic criteria into a single continuous surface, allowing areas where multiple deficits coincide to be identified simultaneously. Unlike the baseline indicators presented in Section 3.1, this output synthesises greenspace accessibility, vegetation condition, deprivation, and child population distribution to provide a more direct indication of where green space interventions are most strongly justified.

Figure 3.3 shows that high suitability values are spatially clustered rather than evenly distributed across Greater Manchester. These clusters are mainly found in the inner-urban areas of Manchester and nearby boroughs, where limited access to green spaces and low vegetation tend to overlap with higher deprivation and more children living in those neighbourhoods. In contrast, lower suitability values are more common in suburban and outer city areas where environmental conditions are generally more favourable.

To support clearer prioritisation, the continuous surface was classified into five suitability categories. As shown in Figure 3.4, areas of very high suitability form coherent neighbourhood-scale zones rather than isolated fragments. Comparison between the continuous and classified outputs demonstrates that classification enhances interpretability without altering the overall spatial pattern, confirming that priority areas reflect broader neighbourhood conditions rather than localised extremes. Together, these outputs highlight the value of an integrated suitability approach for identifying locations where green space interventions are likely to deliver the greatest combined environmental and social benefit.

***Figure 3.3: Continuous composite suitability surface integrating environmental and socio-demographic criteria.***

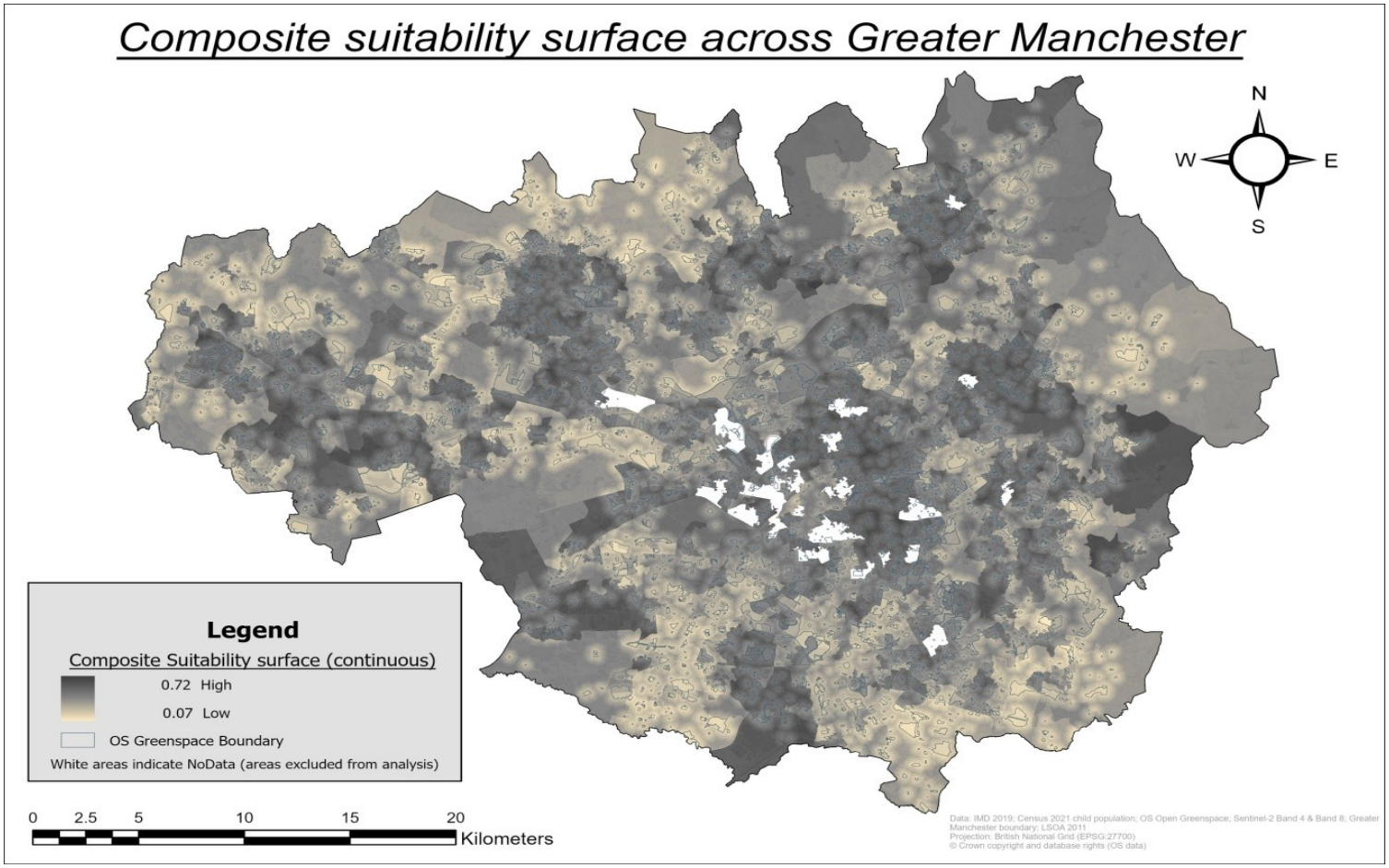
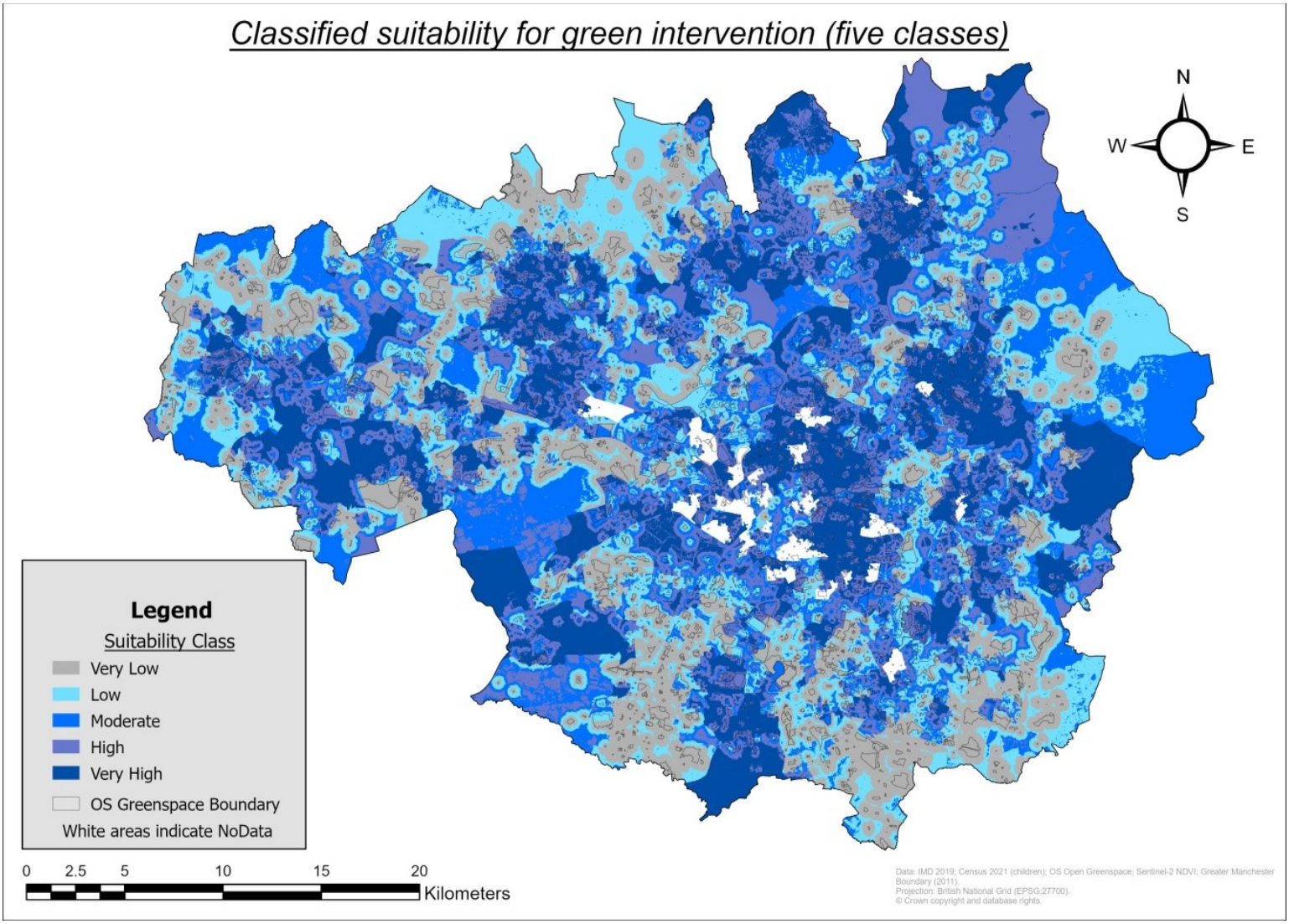


***Figure 3.4: Classified suitability surface showing five levels of intervention priority.***




## 3.3 Priority Areas and Population Impact

The final stage of the analysis turns the composite suitability results into clearly defined priority areas and assesses their potential impact on the population, with a particular emphasis on children. This step moves from identifying spatial patterns of need to isolating neighbourhoods where green space interventions are most likely to deliver meaningful social benefit.

**Figure 3.5** shows the extraction of the highest-priority suitability zones, corresponding to the very high suitability class. These zones tend to group together rather than being scattered, showing that social and environmental inequalities frequently overlap in certain neighbourhoods. This pattern shows how effective the combined approach is, as it helps pinpoint places where improving green spaces could tackle several problems at once.

Suitability values were aggregated to the LSOA level for neighbourhood-scale interpretation. **Figure 3.6** shows mean suitability scores by LSOA, with clear variation across Greater Manchester and higher values clustered in inner-urban neighbourhoods. This aggregation links the raster model outputs to practical policy units, making it easier to discuss priorities in real planning terms.

The relationship between suitability and child population distribution is illustrated in **Figure 3.7**, which highlights LSOAs where high suitability coincides with larger numbers of children. These neighbourhoods are prime spots for green space upgrades, as they'd deliver the biggest social benefits especially for kids, given the higher child populations there who would gain the most. **Table 3.1** backs this up with numbers, summarising mean suitability values by LSOA and confirming the neighbourhood differences you can see in the maps.

Together, the results show that green space needs in Greater Manchester are concentrated in certain areas and closely tied to where more children live. These findings highlight specific neighbourhoods where targeted investment is most justified, directly supporting the research aim and guiding clear recommendations.

***Figure 3.5: Extraction of very high suitability zones representing highest-priority areas for green space intervention.***

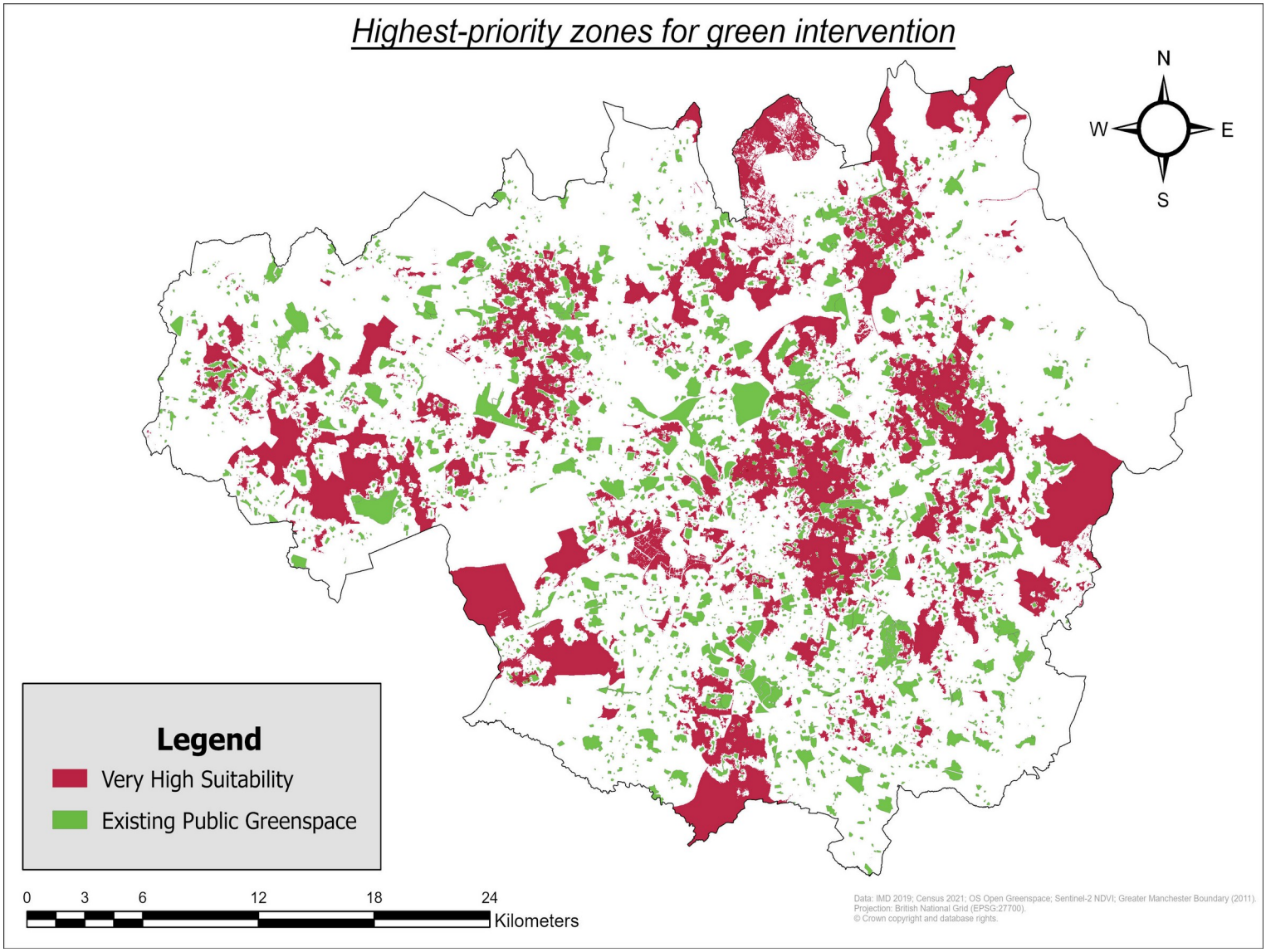


***Figure 3.6: LSOA-level mean composite suitability scores derived from zonal statistics.***

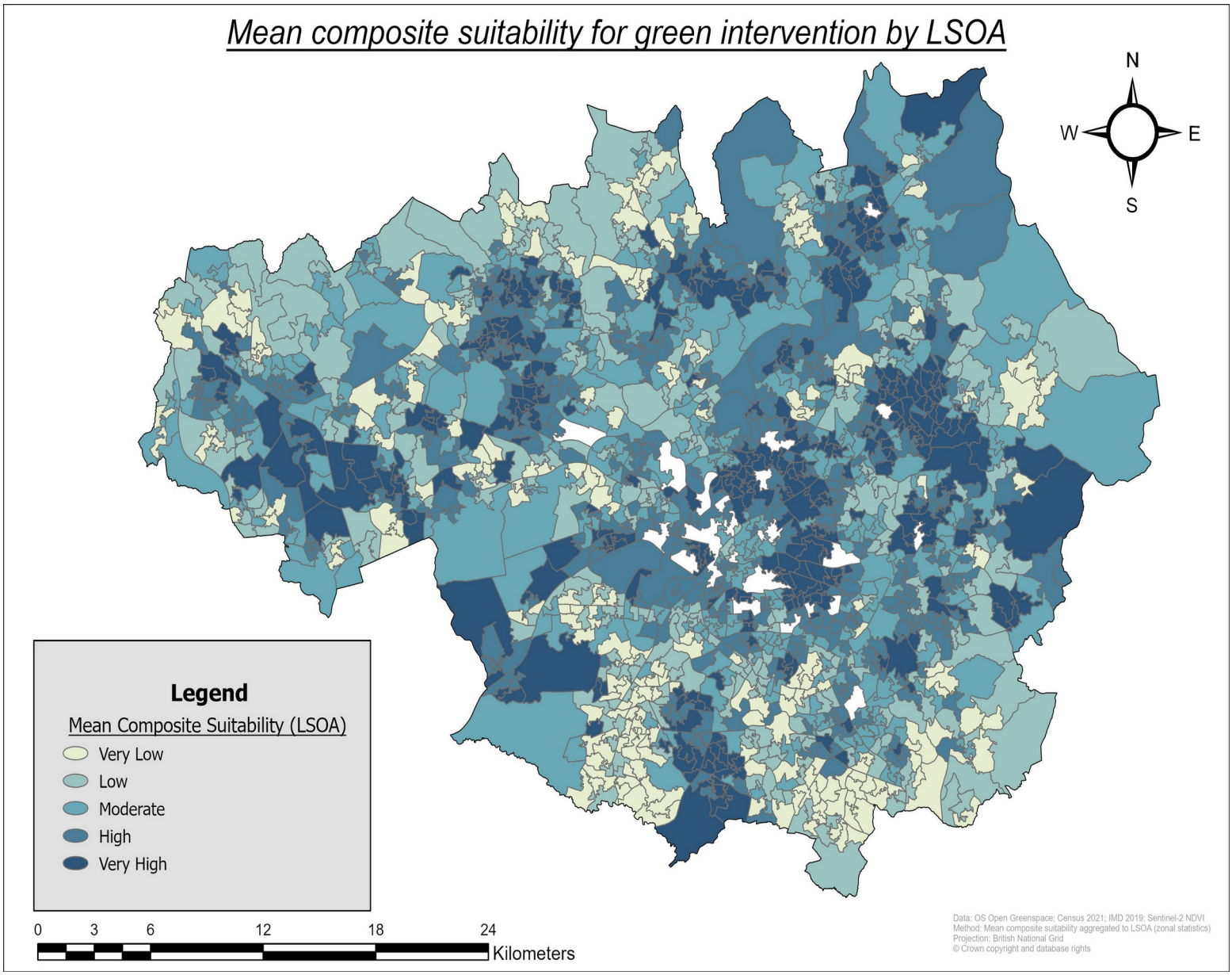


*Figure 3.7: Priority neighbourhoods where high suitability coincides with higher child population.*

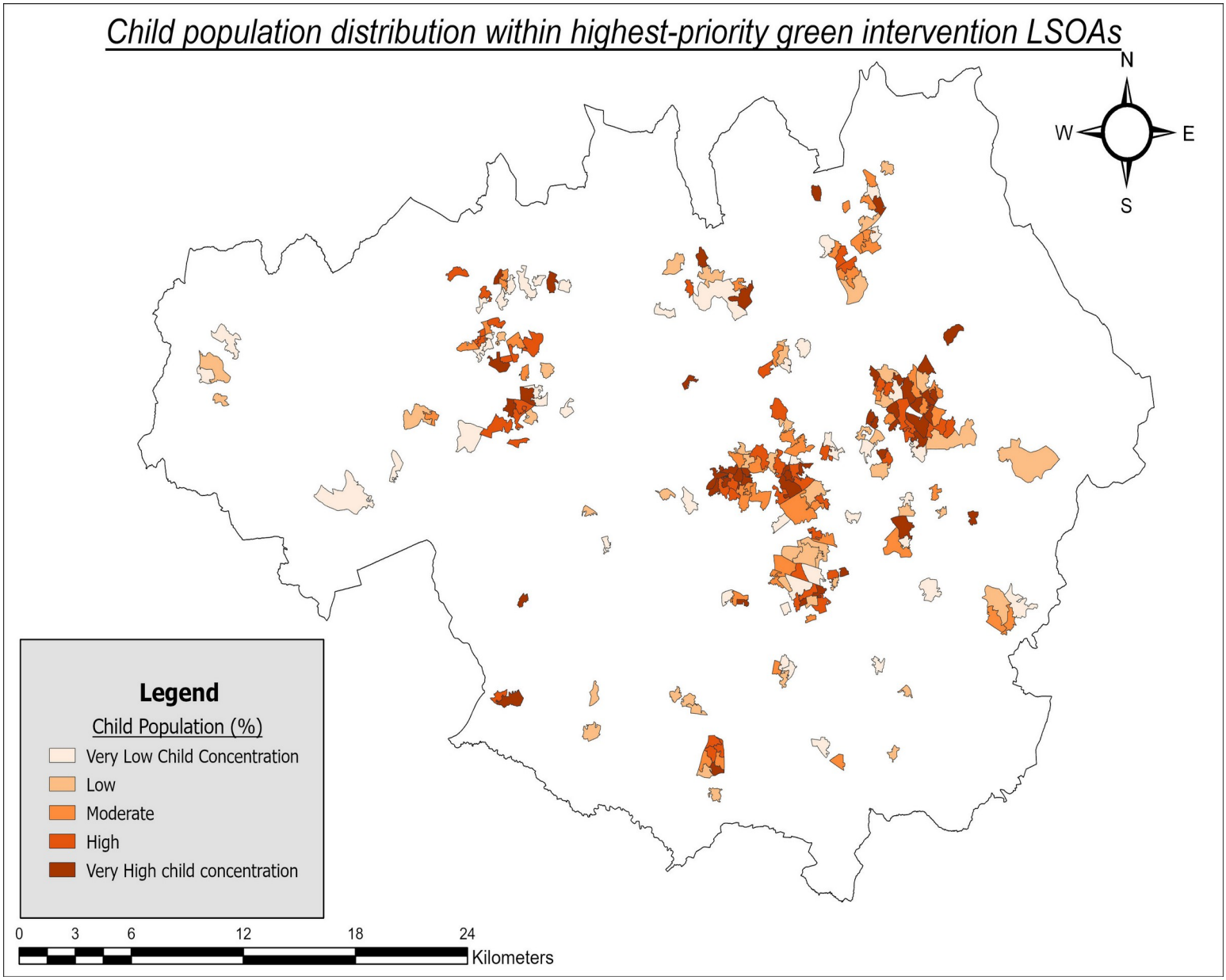



**Table 3.1:** ***Mean composite suitability values by LSOA.***

| Statistic | Mean Suitability |
|---|---|
| Minimum | 0.138 |
| Maximum | 0.637 |
| Mean | 0.395 |
| Median | 0.406 |
| Standard Deviation | 0.110 |

# 4. Limitations And Improvements

This suitability model acts as a screening tool, so the results indicate *relative* priorities rather than exact levels of need. Some data come from different years (for example, IMD 2019 and Census 2021 child data), which is fine for a city-wide analysis but might slightly affect rankings where neighbourhoods have changed rapidly. The greenspace access layer uses straight-line (Euclidean) distance, so it may overestimate access in areas with barriers like railways, rivers, or major roads. NDVI also represents vegetation at one point in time rather than across the whole year.

To improve the model, future work could:
(1) replace Euclidean with network-based walking distance
(2) use multi-date or seasonal NDVI to reduce single-date bias
(3) run a basic sensitivity test on weights and class breaks to assess stability
(4) include feasibility layers such as land ownership, availability, or cost, which weren't covered here.

# 5. Conclusion & Recommendations

This study set out to find neighbourhoods in Greater Manchester where improving green spaces could have the greatest positive impact on children. A GIS-based model was used to combine several datasets, including greenspace access, vegetation cover, deprivation levels, and child population distribution. The results show that environmental disadvantages affecting children are unevenly distributed across the city, with the greatest challenges concentrated in inner-city neighbourhoods.

By bringing these factors together into one suitability model, the study takes a more holistic view, showing where social and environmental challenges overlap the most. Summarising the results at the LSOA level also ties the findings to policy-relevant areas, making it easier to focus on neighbourhoods where action can have the biggest impact. Interestingly, the highlighted priority neighbourhoods aren't simply those with the least green space overall, they are the places where poor access and low vegetation coincide with high child populations and greater socio-economic disadvantage.

From a planning perspective the findings back a child-focused, equity-driven strategy for investing in green infrastructure. Rather than focusing solely on increasing greenspace quantity, targeted fixes, such as improving access to existing parks, enhancing local vegetation cover, or introducing street-level greening around schools and residential streets are likely to  make a bigger social difference in dense urban areas.. While the analysis relies on proxy measures such as distance-based accessibility and NDVI-derived vegetation, and does not capture qualitative aspects of greenspace use or quality, it provides a transparent and defensible framework for prioritisation. Overall, the study demonstrates the value of GIS-based suitability analysis as a decision-support tool for directing green space investment towards neighbourhoods where environmental improvements are most needed and likely to deliver meaningful benefits for children.

# 6. Acknowledgements

This study used OS Open Greenspace data provided by Ordnance Survey.
Contains Ordnance Survey data © Crown copyright and database rights 2024.
Sentinel-2 satellite imagery was obtained from the Copernicus Open Access Hub, and socio-demographic data were sourced from the UK Government and Office for National Statistics. The author also acknowledges the University of Manchester for providing access to ArcGIS Pro and teaching resources used in this analysis.

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# Appendix:

| S.No | Dataset | Source | Scale / resolution | Year | CRS | Purpose in analysis | Source link |
|---|---|---|---|---|---|---|---|
| 1 | LSOA Boundaries (LSOA 2011) | UK Data Service | Vector polygons (LSOA) | 2011 | British National Grid (EPSG: 27700) | Reporting units; join base for IMD & Census | https://census.ukdataservice.ac.uk/get-data/boundary-data.aspx |
| 2 | Index of Multiple Deprivation (IMD) | UK Government | LSOA table | 2019 | – | Core social-equity indicator (weighted in suitability model) | https://www.gov.uk/government/statistics/english-indices-of-deprivation-2019 |
| 3 | Census 2021 Age Structure (TS007A) | Office of National Statistics (Nomis bulk download) | LSOA table | 2021 | – | Child vulnerability indicator (0–14 fields → Child% / Child count) | https://www.nomisweb.co.uk/sources/census_2021_bulk |
| 4 | OS Open Greenspace | Ordnance Survey | Vector | 2024 | British National Grid (EPSG: 27700) | Input for greenspace accessibility (distance/deficit) | https://osdatahub.os.uk |
| 5 | Population Density Grid | OpenPop Grid | Raster (10 m) | 2018 | British National Grid (EPSG: 27700) | Demand/ exposure weighting (and used for raster alignment) | https://openpopgrid.geodata.soton.ac.uk |
| 6 | Sentinel-2 Bands 4 & 8 | ESA Copernicus | Raster (10 m) | 2024 | British National Grid (EPSG: 27700) | NDVI calculation → vegetation deficit indicator | https://dataspace.copernicus.eu |
| 7 | Study area boundary | Derived | Dissolved LSOA polygons | 2024 | British National Grid (EPSG: 27700) | Clip & processing extent | Derived from LSOA 2011 |